

Linear Regression

(Bayesian inference for linear regression.)

$$Y_1, \dots, Y_n \text{ iid } \mathcal{N}(\mu, \sigma^2) \quad \text{univariate.}$$

$$Y_1, \dots, Y_n \text{ iid } \mathcal{N}(X_i^T \beta, \sigma^2) \quad \text{multivariate.}$$

~~Prior distn.~~
Prior predictive distn.
Posterior predictive distn.

MAP

Credible interval

$$\mu \rightarrow \pi(\mu) \propto 1.$$

$$\mu \sim \mathcal{N}(\mu_0, \tau^2)$$

$$\beta \rightarrow \pi(\beta) \propto 1.$$

$$\beta \sim \mathcal{N}(\beta_0, \tau^2)$$

$\beta_j = 0$ vs $\beta_j \neq 0$
Final credible interval.

LASSO / RIDGE

Bayesian formulation equivalent to LASSO / RIDGE.

Missingness

Parameter.

$$Y = X\beta + \epsilon. \quad Y = (Y_1, \dots, Y_n)^T.$$

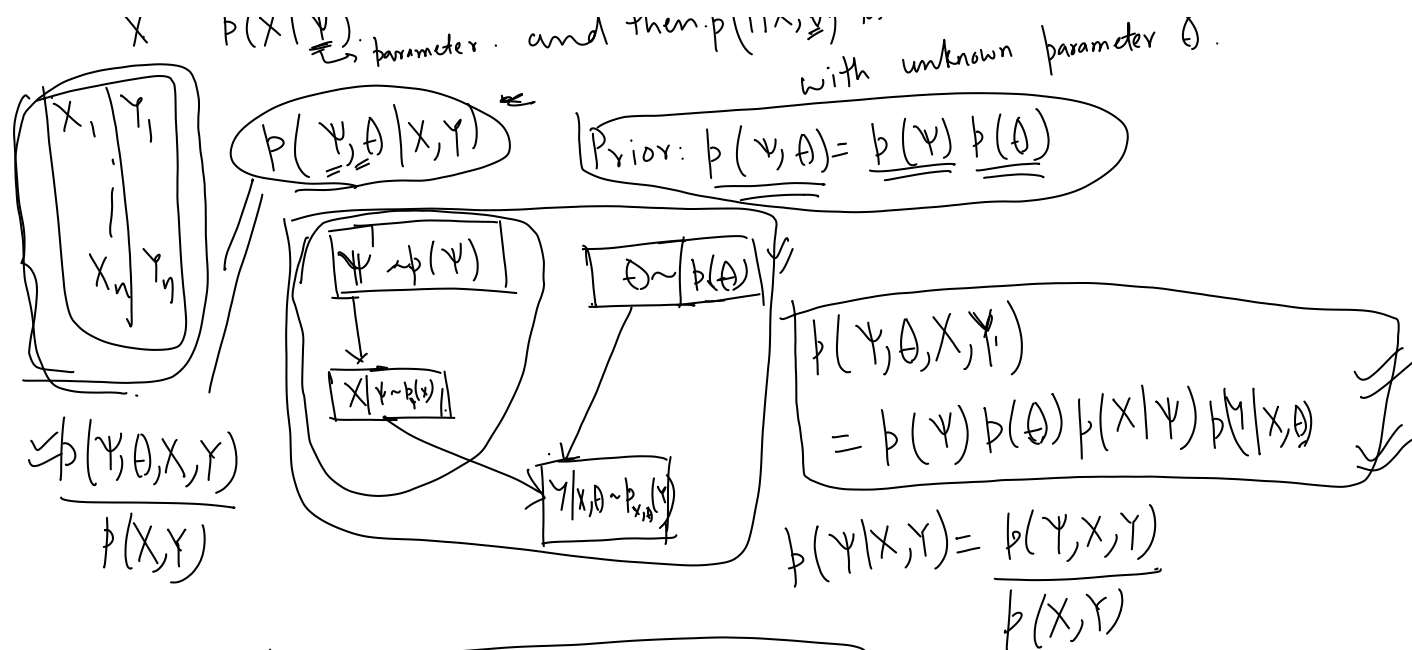
$$\beta = (\beta_0, \beta_1, \dots, \beta_p)^T.$$

Probabilities are essential for Bayesian inference. $\epsilon \sim \mathcal{N}(0, \sigma^2 I_n)$ ✓
We need to have some distn for ϵ as well

$Y|X$ follows some distn (conditional distn. (Regression))

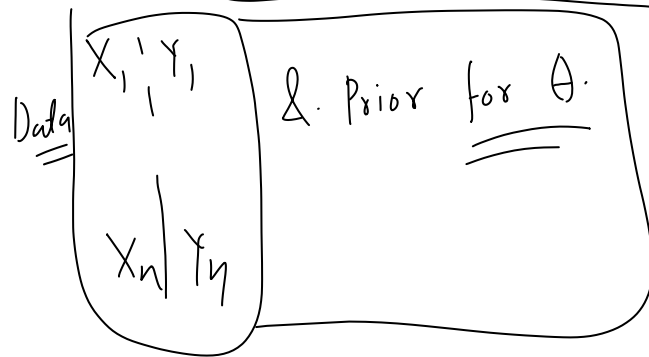
Does the distn of X matter?

... in the regression model.



$$p(\theta | X, Y) \propto p(\theta) p(Y | X, \theta)$$

$$= \frac{\int p(\psi, \theta, X, Y) d\theta}{p(X, Y)}$$



$$Y | X, \theta \sim \text{dist}^n(X, \theta)$$

Show that

$$p(\psi, \theta | X, Y) \propto p(Y | X) p(\theta | X, Y)$$

Prior predictive distⁿ.

$$x^* = \begin{pmatrix} 1 \\ x_1^* \end{pmatrix}$$

B, Σ are independent

$$Y^* | \beta, x^* \sim N(x^{*T} \beta, 1)$$

$$\beta \sim N(0, \tau^2 I)$$

What is the distⁿ of $Y^* | x^*$

$$Y^* = X^{*T} \beta + \epsilon \quad E(Y^* | X^*) = E(X^{*T} \beta + \epsilon) = X^{*T} \beta$$

$$V(X^{*T} \beta + \epsilon | X^*) = \tau^2 X^{*T} X^* + 1$$

$$E(Y^* | X^*) = 0$$

$$V(Y^* | X^*) = \tau^2 X^{*T} X^* + 1$$

$$Y^* | X^* \sim N(0, X^{*T} X^* \tau^2 + 1)$$

$$\beta = (P_0, P_1)^T$$

$$Z: \text{known variance of } \beta$$

$$Y = X \beta + \epsilon$$

$$\begin{matrix} n \times 1 & n \times p & p \times 1 & n \times 1 \end{matrix}$$

$$\epsilon \sim N(0, \sigma^2 I_n)$$

$$\sigma^2 \text{ is known}$$

Then what is the distⁿ of.

$$Y | X \sim \text{MVN}(X \beta_0, X \Sigma_0 X^T + \sigma^2 I)$$

$$\beta \sim N(\beta_0, \Sigma_0)$$

$$\epsilon \text{ \& } \beta \text{ are independent}$$

$$Y = \mu + \epsilon$$

$$\text{corr}(Y_i, Y_j) = \text{corr}(\mu_i, \mu_j)$$

check $\text{corr}(Y_i, Y_j | X) \neq \text{corr}(\mu_i, \mu_j)$ generally.

(specially $\Sigma_0 = I$)

Check what happens when X is orthonormal.

When the condition holds? For what value of n is the correlation highest when $\|x\| = 1$.

$$A \sim N$$

$$B \sim N$$

$$A, B \text{ independent}$$

$$\text{then } A+B$$

$$e^T \Sigma e$$

(σ^2 is known = 1) $Y = XB + E$. $E \sim N(0, \sigma^2 I_n)$.

$$L(\beta; y) \propto \exp \left\{ -\frac{1}{2} (Y - XB)^T (Y - XB) \right\}$$

$\frac{n^2 - 2na}{(n-1)^2}$

$$\exp \left\{ -\frac{1}{2} \frac{\sum (y_i - \bar{y})^2}{\sigma^2} \right\}$$

$$\exp \left\{ -\frac{1}{2} \sum (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2 \right\}$$

$$\propto \exp \left\{ -\frac{1}{2} n(\bar{y} - \mu)^2 \right\}$$

$$= \exp \left\{ -\frac{1}{2} (Y^T Y + B^T X^T X B - 2 B^T X^T Y) \right\}$$

$$\propto \exp \left\{ -\frac{1}{2} (B^T X^T X B - 2 B^T X^T Y) \right\}$$

$$\exp \left\{ -\frac{1}{2} (B - B_0)^T \Sigma^{-1} (B - B_0) \right\} = \exp \left\{ -\frac{1}{2} B^T \Sigma^{-1} B - 2 B_0^T \Sigma^{-1} B \right\}$$

$$\exp \left\{ -\frac{1}{2} B^T A B - 2 B^T c \right\}$$

$$= \exp \left\{ -\frac{1}{2} (B - \alpha_0)^T W (B - \alpha_0) \right\}$$

Suppose M is symmetric & invertible.

$$x^T M x - 2 b^T x$$

$$\left(x - \frac{b}{\sigma^2} \right)^T \frac{1}{\sigma^2} \left(x - \frac{b}{\sigma^2} \right)$$

$$(x - c)^T M (x - c) = x^T M x - 2 c^T M x + c^T M c.$$

$$\Rightarrow b^T = c^T M$$

$$\Rightarrow M c = b \Rightarrow \boxed{c = M^{-1} b}$$

$$c^T M c = b^T M^{-1} b$$

$$= \frac{1}{n} \sum_{i=1}^n (x_i - M^{-1} b)^T (x_i - M^{-1} b) = b^T M^{-1} b$$

$$\Rightarrow \underline{X^T M x - 2b^T x = (x - M^{-1}b)^T M^{-1}b}$$

$$\beta \sim N(0, \tau^2 I) \checkmark$$

$$\text{Then, } \beta | Y \sim N \left(\underline{(X^T X + \tau^2 I)^{-1} (X^T Y)}, (X^T X + \tau^2 I)^{-1} \right)$$

$$\left(\begin{pmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{pmatrix} \right) \left(\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \right)$$

$$\exp \left\{ -\frac{1}{2} \left(\underline{Y^T Y + \beta^T X^T X \beta - 2 \beta^T X^T Y} \right) \right\} \exp \left\{ -\frac{1}{2} \frac{\beta^T \beta}{\tau^2} \right\}$$

$$\cancel{\tau^2} \tau \rightarrow \infty \quad \left((X^T X)^{-1} X^T Y, (X^T X)^{-1} \right)$$

$$\tau \rightarrow 0 \quad \text{for } \beta | Y \approx \text{Point mass at } 0$$

$$x^T (\underline{X^T X} + \tau^2 I) x$$

$$x^T (X^T X) x \geq 0$$

$$x^T (\tau^2 I) x > 0$$

$$(X^T X)^{-1} X^T Y$$

$$X^T X +$$

$$\text{MAP is } (X^T X + \tau^2 I)^{-1} X^T Y$$

σ^2 is unknown.

$$L = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp \left\{ -\frac{(y - X\beta)^T (y - X\beta)}{2\sigma^2} \right\}$$

$$h(\sigma^2) \propto (\sigma^2)^{-(a_0+1)} \exp \left(-\frac{b_0}{\sigma^2} \right) = I h(a_0, b_0)$$

$$p(\beta, \sigma^2) = \underbrace{p(\sigma^2)}_{IG} \underbrace{p(\beta | \sigma^2)}_{MVN} \quad \& \quad \beta | \sigma^2 \sim N(\beta_0, \sigma^2 V_0)$$

Do the calculation for the case with $\beta_0 = 0, V_0 = I$

$$p(\beta, \sigma^2 | y)$$

$$p(\sigma^2 | y) = \int p(\beta, \sigma^2 | y) d\beta$$

Sample: $p(\sigma^2 | y)$, then sample $p(\beta | \sigma^2, y)$ $\sigma^2 | y \sim IG(a_n, b_n)$ where $a_n = a_0 + \frac{n}{2}$
 $b_n = b_0 + \frac{1}{2} [y^T y - \beta_n^T V_n^{-1} y]$

Prior: $p(\beta, \sigma^2) = p(\sigma^2) p(\beta | \sigma^2)$

where $V_n = (X^T X + I)^{-1}$
 $\& \beta_n = V_n X^T y$

$$p(\beta, \sigma^2 | y) = \underbrace{p(\sigma^2 | y)}_{IG} \underbrace{p(\beta | \sigma^2, y)}_{MVN}$$

(conjugate!)

Gibbs sampler:

$$p(\beta, \sigma^2 | y)$$

Initialise $\sigma^{2(0)}$

$$p(\beta | \sigma^{2(0)}, y)$$

Initial $Y_0, X | Y$

(MNN)

β^0

$$p(\underline{\sigma^2} | \underline{\beta}, y) \propto p(\sigma^2, \beta | y) \cancel{p(\beta)}$$

$$p(a | b, c, y) \propto p(a, b, c, y) = g(a) \cancel{h(b)}$$

$\propto \underline{g(a)}$

$$p(\sigma^2 | \beta, y) \propto \frac{p(\sigma^2)}{p(y)}$$

$$= (\sigma^2)^{-n/2} \exp$$

$$\frac{1}{2} (y - \beta)^T (y - \beta) + \frac{1}{2}$$

$$(\sigma^2)^{-(a+1)} \cdot (\sigma^2)^{b+1}$$

$$= (\sigma^2)^{n/2 + a + 1 + b}$$

